

Utkarsh Sonawane

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EDUCATION

The Pennsylvania State University

B.S. Computer Science | Capital Honors Program

Harrisburg, PA

Expected Dec 2028

– **GPA:** 3.53/4.00 | Dean's List

– **Coursework:** Data Structures & Algorithms; Object-Oriented Programming and Calculus III (in progress)

TECHNICAL SKILLS

Languages: Python, Swift, JavaScript, TypeScript, SQL, Java

Backend & Data: FastAPI, Pydantic, PostgreSQL (row-level security), Supabase, pgvector, Node.js, Express, REST APIs

Apple & Web: SwiftUI, HealthKit, WatchConnectivity, Apple Foundation Models, React

Testing & Tools: pytest, XCTest, Vitest, mypy, GitHub Actions, Git, Linux

PROJECTS

Nytr — Nutrition & Training Planner | *Python, FastAPI, PostgreSQL, SwiftUI*

2026

- Built an iOS app and FastAPI backend that plan meals deterministically from Penn State Harrisburg's daily dining menu, combining HealthKit body and workout data, Hevy training history, and barcode-scanned foods.
- Modeled all data as immutable, provenance-tracked evidence: versioned foods, targets, and menu snapshots, append-only corrections, exact Decimal arithmetic, Postgres row-level security, and fail-closed handling of stale data.
- Kept on-device AI (Apple Foundation Models) strictly downstream of computed results; 1,135 backend tests (1,124 passing in CI against Postgres) under strict mypy, plus 251 iOS XCTest cases.

Talks — Apple Watch Meeting Recorder | *Swift, SwiftUI, watchOS*

2026

- Built a watchOS/iOS app that records on the Watch, transfers audio via WatchConnectivity, transcribes on device (SpeechAnalyzer, SFSpeechRecognizer fallback), summarizes with Apple Foundation Models, and writes to Notion.
- Made delivery durable: ACK-gated deletion on the Watch, a persistent job queue that survives app termination and reboot, and idempotent Notion retries that append only missing blocks; 57 XCTest cases passing in CI.

Clage — NEAT Engine & Artificial Life | *Python, matplotlib*

2026

- Implemented NEAT from scratch (innovation tracking, crossover, speciation); over 5 seeded trials each it solves OR and AND 5/5 and XOR 4/5; sine is not solved within 300 generations.
- Built a 2D artificial-life world and a seeded experiment harness that changes one environmental factor at a time against a control, with behavioral metrics and replay tools; 300 tests, passing in CI.

Devvy — Local Discord Presence Daemon | *JavaScript, Node.js, Discord IPC*

2026

- Built a Node.js daemon that merges VS Code, OpenCode, and Command Code activity into one Discord presence, using priority arbitration, 15-second source expiry, binary Discord IPC framing, and reconnect with backoff.
- Enforced a privacy boundary (integrations send only project, mode, and model over loopback; never prompts, code, or paths) and gated macOS releases on CI; 26 test scripts, all run in CI, incl. a mock Discord IPC server.

LEADERSHIP & ACTIVITIES

IEEE Student Branch, Penn State Harrisburg

Harrisburg, PA

Technical Chair (Sep 2026 – Present); Technical Team Member (Sep 2025 – Sep 2026)

Sep 2025 – Present

- Elected Technical Chair in Sep 2026; coordinate software, AI, and robotics workshops and projects for the branch.
- Hosted 2 hands-on technical workshops for students: Markdown and building with Google Gemini.

Founder & Editor-in-Chief

Dubai, UAE

Glimpse — Student Publication

Jun 2024 – Feb 2025

- Grew a new student publication to 250+ subscribers; wrote and edited 30+ articles on technology, science, and student life.

HONORS & CERTIFICATIONS

USAII Global AI Hackathon 2026: Finalist team (5 members) among 6,081+ participants; built ScholarAI's data pipeline (normalization, embeddings, pgvector).

Certifications: Machine Learning Specialization (*DeepLearning.AI*) | Practical Deep Learning for Coders (*fast.ai*) | Ethics of AI (*University of Helsinki*) | AI Essentials (*Penn State*)